Coordinate Geometry: Circles

Questions 1–51

Question 1. State the centre and radius of the following circles:

a) $(x-2)^2+(y-7)^2=3^2$
b) $x^2+(y+2)^2=3.5^2$
c) $(x+8)^2+(y-5)^2=16$
d) $(x+12)^2+(y+9)^2=19$
e) $(x-3)^2+(y+16)^2=28$
f) $(x-\frac{1}{2})^2+y^2=\frac{9}{4}$
g) $(x+2k)^2+(y-k+1)^2=100k,\ k>0$

Question 2. Write down the equation of the circle in the form $(x+a)^2+(y+b)^2=c$, where $c>0$, given the following centre and radius:

a) Centre $(-3,-8)$, Radius 9
b) Centre $(4,-3)$, Radius 5
c) Centre $(0,0)$, Radius 100
d) Centre $(6,1)$, Radius $\sqrt{17}$
e) Centre $(-11,6)$, Radius $2\sqrt{2}$
f) Centre $(-\frac{2}{3},-\frac{3}{2})$, Radius $\frac{4}{9}$
g) Centre $(k-5,7-k)$, Radius $(k+2),\ k>0$

Question 3. Write down the exact area of these circles:

a) $(x+4)^2+(y-1)^2=5^2$
b) $x^2+(y+\frac{3}{8})^2=\frac{49}{9}$
c) $3(x+7)^2+3y^2=363$

Question 4. Write the equations for each of these circles in the form $x^2+ax+y^2+by+c=0$:

a) $x^2+(y+6)^2=2$
b) $(x-10)^2+y^2=130$
c) $(x+2)^2+(y-8)^2=66$
d) $(x-11)^2+(y-15)^2=1600$
e) $(x+4k)^2+(y-2k)^2=20k^2$

Question 5. Write the equations for each of these circles in the form $(x+a)^2+(y+b)^2=c$, where $c>0$:

a) $x^2+2x+y^2+6y-8=0$
b) $x^2-8x+y^2-20y-784=0$
c) $x^2+y^2+6x-14y-82=0$
d) $x^2+y^2-5x-9y-\frac{57}{2}=0$
e) $x^2-\frac{2}{3}x+y^2+\frac{4}{9}y-\frac{230}{81}=0$
f) $x^2+y^2-\frac{3}{2}x-\frac{5}{4}y-\frac{1091}{64}=0$
g) $x^2+y^2+\frac{14}{5}x+\frac{5}{2}y+\frac{1009}{400}=0$
h) $x^2+y^2+kx-6ky+3k=0,\ k>\frac{12}{37}$

Question 6. Determine whether the following equations define a circle:

a) $x^2-6x+y^2+16y+83=0$
b) $x^2+10x+y^2-y+\frac{101}{4}=0$
c) $x^2-4x-y^2-10y-46=0$
d) $x^2+12x+3y^2-6y+3=0$
e) $x^2+\frac{2}{3}x+y^2-\frac{4}{3}y-\frac{1}{3}=0$

Question 7. Determine the possible values of k such that the equation $x^2 + y^2 + 8x - 12y - k = 0$ represents a circle.

Question 8. Determine the possible values of k such that the equation $x^2 + y^2 - 3x + 9y + \frac{k}{6} = 0$ represents a circle.

Question 9. Determine whether these circles intersect with either the x-axis or the y-axis:

a) $(x-4)^2+(y-5)^2=9$
b) $(x+1)^2+(y-6)^2=25$
c) $(x+3)^2+(y-4)^2=8$
d) $(x+8)^2+(y+3)^2=10$

Question 10. By considering the centre and radius, determine the values of k, where $k>0$, such that these circles intersect the x-axis at two distinct points:

a) $(x-2)^2+(y-k)^2=36$
b) $(x+5)^2+(y-k)^2=10$
c) $(x-1)^2+(y-4)^2=k^2$
d) $(x+7)^2+(y+8)^2=(k+1)^2$

Question 11. By considering the centre and radius, determine the values of k, where $k > 0$, such that these circles intersect the y-axis at two distinct points:

a) $(x-k)^2 + y^2 = 49$
b) $(x+k)^2 + (y-4)^2 = 9$
c) $(x-6)^2 + (y+3)^2 = k^2$

Question 12. Sketch the following circles, identifying their centre and the coordinates of the point on the circle that is closest to the x-axis:

a) $(x-3)^2 + (y+5)^2 = 4$
b) $(x-6)^2 + (y-3)^2 = 4$
c) $(x+7)^2 + (y+5)^2 = 16$
d) $(x+6)^2 + (y-6)^2 = 10$

Question 13. Sketch the following circles, identifying their centre and where they intersect the coordinate axes:

a) $x^2 + (y-3)^2 = 9$
b) $x^2 + (y+2)^2 = 16$
c) $(x-5)^2 + y^2 = 81$
d) $(x+5)^2 + y^2 = 25$

Question 14. Sketch the following circles, identifying their centre and where they cross the coordinate axes:

a) $(x-4)^2 + (y-4)^2 = 16$
b) $(x+2)^2 + (y+2)^2 = 13$
c) $(x-1)^2 + (y+3)^2 = 17$
d) $(x+7)^2 + (y-2)^2 = 20$

Question 15. Sketch the following circles, identifying their centre and the exact points at which they cross the coordinate axes:

a) $(x+3)^2+(y-8)^2=16$
b) $(x+7)^2+(y+5)^2=36$
c) $(x-8)^2+(y-2)^2=81$
d) $(x-1)^2+(y-\frac{11}{16})^2=\frac{697}{256}$

Question 16. Determine whether the point $(3,7)$ is inside, on, or outside of these circles:

a) $(x-1)^2+(y-5)^2=9$
b) $(x-3)^2+(y-9)^2=4$
c) $(x-5)^2+(y-6)^2=1$

Question 17. A circle has centre $(2, -2)$ and the point $(14, 14)$ lies on the circle. Find the equation of the circle.

Question 18. The points A and B lie on a circle. The line AB passes through the centre of the circle. Find the equation of the circle in each case:

a) $A = (-5, -7)$ and $B = (9, -5)$
b) $A = (-8, 7)$ and $B = (6, -1)$
c) $A = (4, 7)$ and $B = (-4, -9)$
d) $A = (-2, -1)$ and $B = (12, 13)$

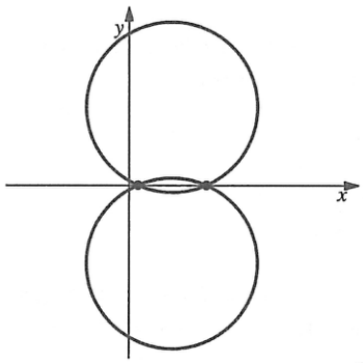
Question 19. A circle passes through the points $A(6, 2)$ and $B(8, 4)$, and the centre of the circle lies on the x-axis. Find the equation of the circle.

Question 20. A circle passes through the points $A(-5, 1)$ and $B(3, 11)$, and the centre of the circle lies on the y-axis. Find the equation of the circle.

Question 21. A circle passes through the points $A(1, -6)$ and $B(17, -10)$, and the centre of the circle lies on the line $y = x$. Find the equation of the circle.

Question 22. Show that the shortest distance from $A(-3, 8)$ to the circumference of the circle with equation $(x - 8)^2 + (y + 3)^2 = 9$ is $a\sqrt{2} + b$, where a and b are integers.

Question 23. The diagram below shows two circles of radius 10 that pass through the points $(1, 0)$ and $(9, 0)$. Find the equations of the two circles.



Question 24. A circle passes through the points $A(7, 2)$, $B(3, 2)$ and $C(3, -6)$. Find the equation of the circle.

Question 25. A circle passes through the points $A(-1,-12)$, $B(0,-13)$ and $C(3,-4)$. Find the equation of the circle.

Question 26. A circle passes through the points $A(-8,1)$, $B(-9,2)$ and $C(0,11)$. Find the equation of the circle.

Question 27. A circle passes through the points $A(1,1)$ and $B(2,6)$. A diameter of the circle lies on the y-axis. Find the equation of the circle.

Question 28. The points $A(-6,-1)$, $B(2,7)$ and $C(4,5)$ lie on a circle. It is given that AC is a diameter of the circle. Find the area of the triangle ABC.

Question 29. A circle passes through the points $A(-12, 11)$, $B(-8, -1)$ and $C(-2, 1)$

a) Calculate the lengths of the sides of the triangle ABC.

b) Hence show that AC is a diameter of the circle.

c) Find the area of triangle ABC.

Question 30. The points $A(2, 6)$ and $B(6, k)$ lie on a circle such that AB is a diameter of the circle. The line through A and B also passes through $(10, 20)$. Find the equation of the circle.

Question 31. Determine whether the following lines intersect the circle with equation $(x+4)^2+(y-6)^2=20$ once, twice or not at all:

a) $y=5x+1$
b) $y=1-4x$
c) $y=2x+24$

Question 32. The line with equation $y=k$ is a tangent to the circle $x^2+y^2-3x+10y+\frac{427}{16}=0$. Find the possible values of k.

Question 33. The vertices of an equilateral triangle lie at $(0, 0)$, $(-10, 10\sqrt{3})$ and $(10, 10\sqrt{3})$. A circle is drawn within the triangle so that the three sides are all tangents to the circle. Find the area of the circle.

Question 34. The circle with equation $(x-3)^2 + (y+5)^2 = 34$ passes through the origin. Find the gradient of the tangent to the circle at the origin.

Question 35. The point $A(1, 2)$ lies on the circle with equation $(x + 2)^2 + (y - 3)^2 = 10$. Find the equation of the tangent to the circle at A in the form $y = ax + b$, where a and b are integers.

Question 36. The point A has coordinates $(\frac{5}{2}, -\frac{1}{2})$ and lies on the circle with equation $(x - \frac{1}{2})^2 + (y + \frac{3}{2})^2 = 5$. Find the equation of the tangent to the circle at A in the form $y = ax + b$.

Question 37. The point A has coordinates $(1, 1)$ and lies outside of the circle $(x + 3)^2 + (y - 9)^2 = 10$. The point B lies on the circle such that AB is a tangent to the circle. Find the length of AB.

Question 38. Find the two equations of the tangents to the circle $(x - 3)^2 + (y - 4)^2 = 100$ when $x = -5$ in the form $y = ax + b$, where a and b are rational numbers.

Question 39. Find the two equations of the tangents to the circle $(x+5)^2+(y-9)^2=10$ when $y=10$ in the form $y=ax+b$, where a and b are integers.

Question 40. The line $x+7y=82$ is a tangent to a circle with centre $(-\frac{5}{2},\frac{17}{2})$. Find the equation of the circle.

Question 41. A circle has equation $(x-11)^2+(y+4)^2=16$. The straight line with equation $y=kx$, where k is a constant, meets the circle.

a) Show that the coordinates of any points of intersection between the line and the circle satisfies the equation $(1+k^2)x^2+2(4k-11)x+121=0$

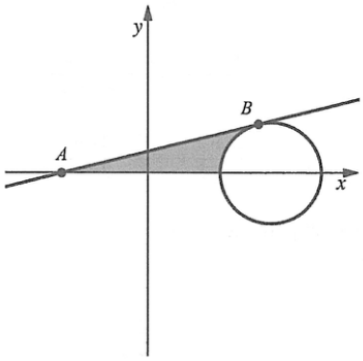
b) Hence, find the two values of k for which the line $y=kx$ is a tangent to the circle.

Question 42. A circle has equation $(x+3)^2+(y-3)^2=80$. The straight line with equation $y=2x+k$, where k is a constant, meets the circle.

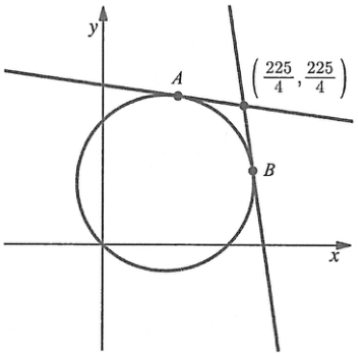
a) Show that the coordinates of any points of intersection between the line and the circle satisfies the equation $5x^2+2(2k-3)x+k^2-6k-62=0$

b) Hence, find the two values of k for which the line $y=2x+k$ is a tangent to the circle.

Question 43. The diagram below shows the circle with equation $(x-10)^2+y^2=17$. The line passing through A and B is a tangent to the circle at the point where $x=9$. Find the area of the shaded region.



Question 44. The diagram below shows the circle with equation $(x-25)^2+(y-25)^2=1250$. The two lines are both tangents to the circle. The tangent at A has equation $y=-\frac{1}{7}x+a$ and the tangent at B has equation $y=-7x+b$. Show that the two lines intersect at $(\frac{225}{4},\frac{225}{4})$.



Question 45. Find the exact shortest distance from the line $y = 2x + 1$ to the circle $x^2 + (y - 7)^2 = 3$.

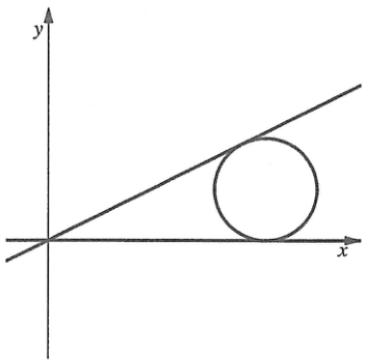
Question 46. A circle has equation $x^2 - 14x + y^2 + 18y - 40 = 0$. The points A and B lie on the circle such that $(2, 1)$ is the midpoint of A and B. Find the shortest distance around the circumference of the circle from A to B.

Question 47. A circle, P, has equation $x^2 + y^2 + 6y = 7$. A second circle, Q, has its centre on the x-axis and has the same radius as P. Q touches P at one point. Find the possible equations of Q.

Question 48. A circle centred at $(8, k)$ has $3x - 2y = 12$ and $3x - 2y = 24$ as tangents. Find the equation of the circle.

Question 49. A circle with radius k has both the x-axis and the y-axis as tangent lines. $2x + y = 50$ is also a tangent line to the circle. Find the two possible exact values of k and sketch both situations on the same graph.

Question 50. A circle of area 25π square units is drawn that has both the x-axis and $y = \frac{1}{2}x$ as tangents, as shown in the diagram below. Find the equation of the circle.



Question 51. For the circle shown in the diagram below, determine the percentage of the circle's area that is below the x-axis.

